Decomposition of Concentration-Index Trees

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1 4. Results

The results section reports only the objects needed to answer the main technical question: whether inequality-oriented recursive partitioning identifies subgroup structure that is visible, stable, and comparable with classical decomposition output. Full generated outputs are moved to the appendix.

1.1 4.1 Exploratory Data Analysis

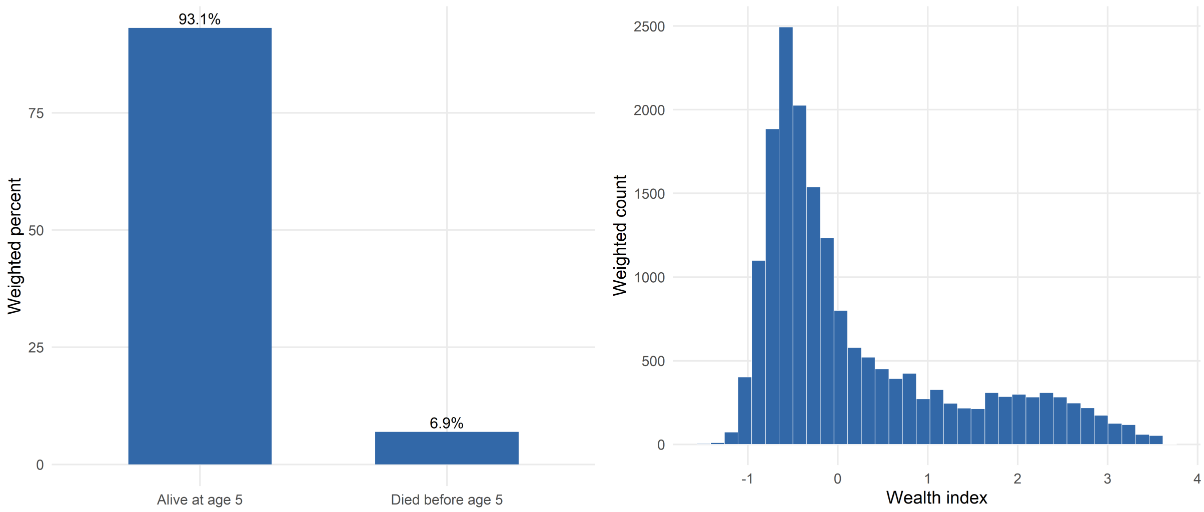

Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 4.2 Application of developed tree models to the DRC DHS Data

1.2.1 4.2.1 Comparison of Indirect Decomposition Results

Categorical predictors enter the regression decomposition as separate one-hot terms. The predictive ranger is fitted on the same design-matrix terms, so the comparison table reports categorical levels at the same granularity instead of collapsing them back to their parent variable.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion

	Regression			Predictive-forest SHAP			
Variable	**CI (%)**	**CIg (%)**	**CIc (%)**	**CI (%)**	**CIg (%)**	**CIc (%)**	**L (%)**
Residence	33.993	70.954	70.954	32.349	32.349	32.349	38.158

Province: Lualaba	6.713	1.192	1.192	-5.373	-5.373	-5.373	-4.558
Province: Sankuru	6.700	1.570	1.570	3.670	3.670	3.670	1.469
Partner occupation: d Agriculture	6.406	7.369	7.369	7.770	7.770	7.770	9.227
Province: Sud-Kivu	4.471	0.665	0.665	5.204	5.204	5.204	3.892
Birth group: c 2-4 long	3.506	3.503	3.503	-5.820	-5.820	-5.820	-4.951
Mother occupation: d Agriculture	3.463	4.521	4.521	15.488	15.488	15.488	15.615
Province: Kwilu	3.342	0.326	0.326	-0.765	-0.765	-0.765	-1.395
Province: Nord-Kivu	2.859	0.243	0.243	-0.323	-0.323	-0.323	-0.002
Partner education	2.750	0.684	0.684	-0.006	-0.006	-0.006	1.724

1.2.2 4.2.2 Root-Node Inequality

Table 2: Root-node inequality in under-five mortality by concentration-index criterion

Root node	**CI**	**CIg**	**CIc**	**L**
Full sample	0.1112	0.0077	0.0308	0.0818

1.2.3 4.2.3 Selected Concentration-Index Tree

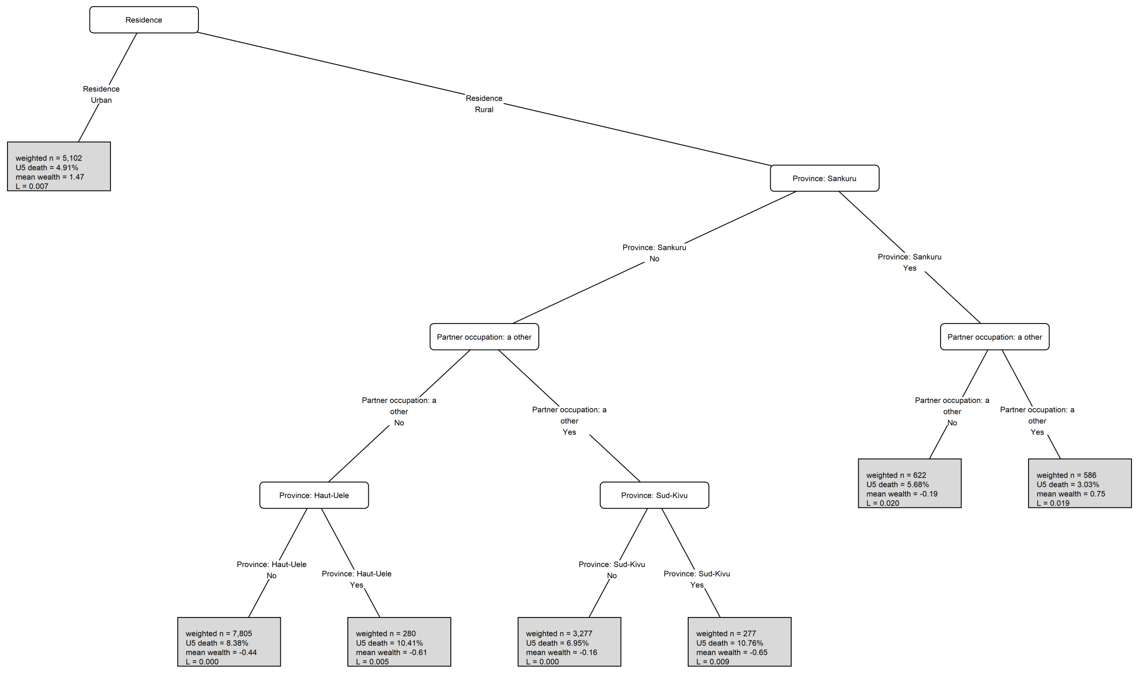

Figure 2: Selected concentration-index tree for under-five mortality in the DRC.

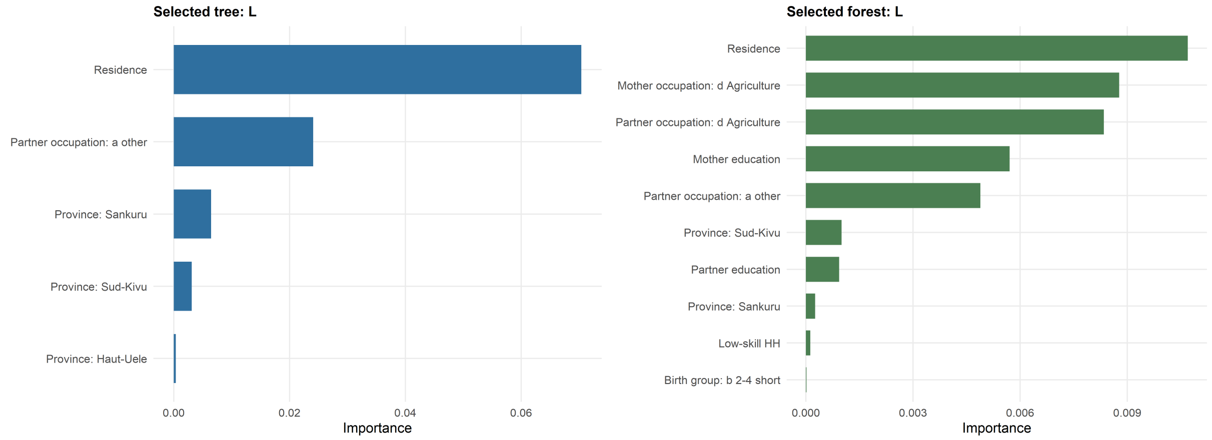

Figure 3: Variable importance for the selected concentration-index tree and forest.

1.2.4 Surrogate Tree Comparison

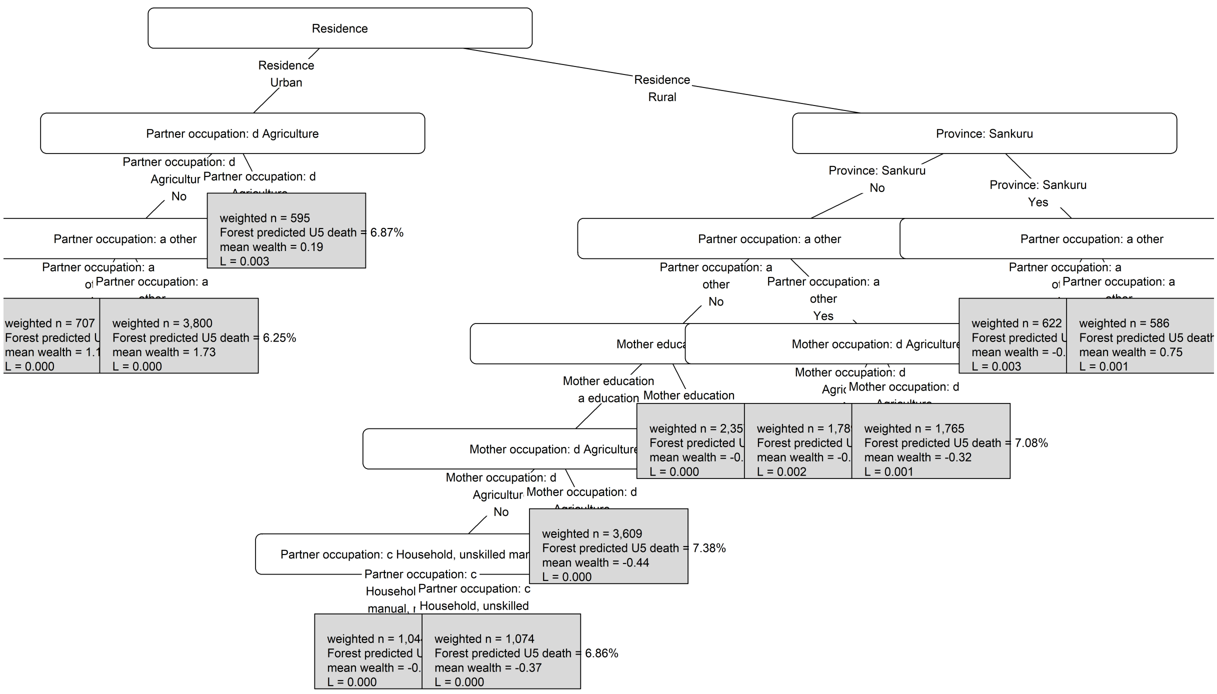

Figure 4: Surrogate tree summarising the selected concentration-index forest predictions.

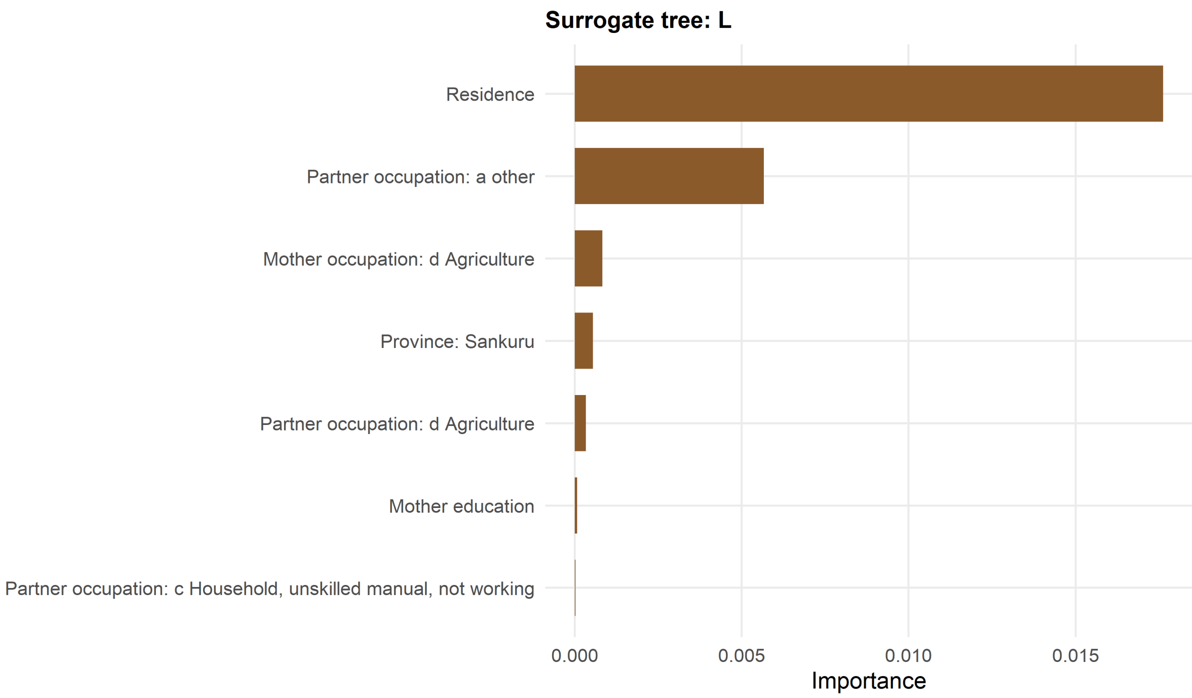

Figure 5: Variable importance for the surrogate tree of the selected concentration-index forest.

1.3 4.3 Simulation Study

1.3.1 4.3.1 Recovery of the Simulated Subgroup Structure

Table 3: Positive-control validation of recovery of the planted subgroup

max_terminal_planted_share	recovered_pure_planted_node	planted_variables_used	all_planted_varia
1	TRUE	rural, ed, reg	

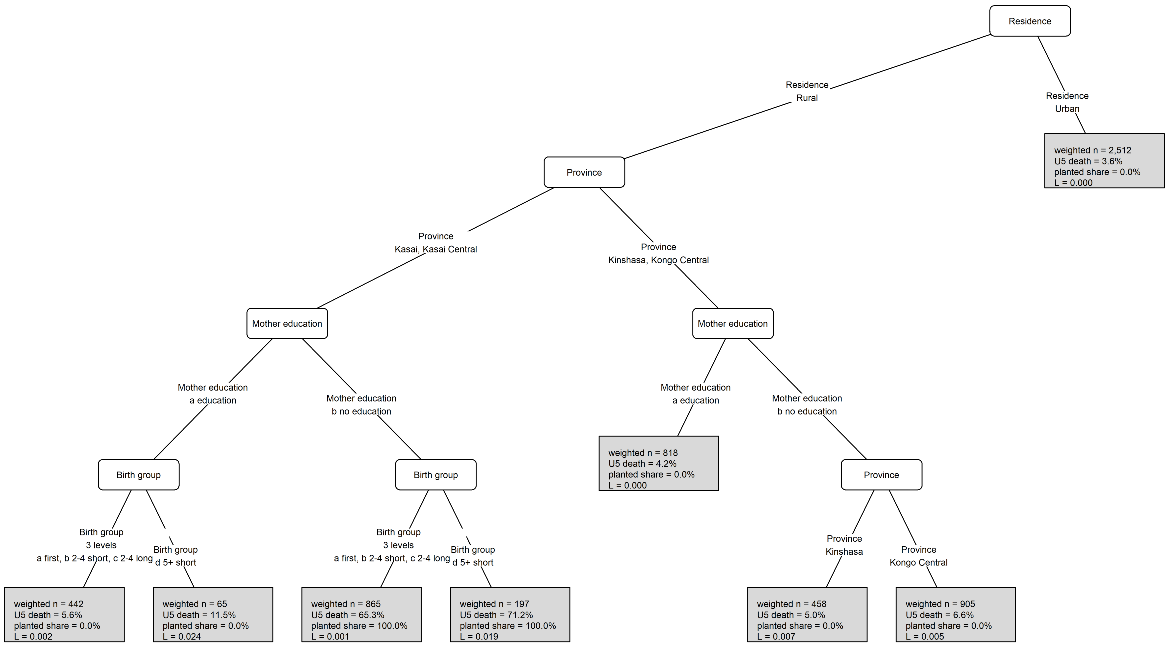

Figure 6: Tree recovered from simulated data with a known subgroup-generating mechanism.

2 Appendix

2.1 Results

2.1.1 Appendix R1: Exploratory Data Outputs

Table 4: Weighted distribution of analysis variables by under-five mortality status

	Under-five mortality		**Overall**
Variable	**Alive at age 5**	**Died before age 5**	**Overall**
	(weighted N=16,704)	(weighted N=1,244)	(weighted N=17,948)
Low-skill HH			
No	10,185.6 (61.0%)	786.1 (63.2%)	10,971.7 (61.1%)
Yes	6,518.4 (39.0%)	458.2 (36.8%)	6,976.5 (38.9%)
Child sex			
Female	8,099.9 (48.5%)	515.2 (41.4%)	8,615.1 (48.0%)
Male	8,604.1 (51.5%)	729 (58.6%)	9,333.1 (52.0%)
Birth group			
a first	2,691.8 (16.1%)	242.1 (19.5%)	2,933.9 (16.3%)
b 2-4 short	2,112.9 (12.6%)	186.1 (15.0%)	2,299 (12.8%)
c 2-4 long	5,861.4 (35.1%)	287 (23.1%)	6,148.4 (34.3%)
d 5+ short	1,633.8 (9.8%)	248.3 (20.0%)	1,882.1 (10.5%)
e 5+ long	4,404.1 (26.4%)	280.7 (22.6%)	4,684.8 (26.1%)
Mother age			
a20 or more	16,100.6 (96.4%)	1,172.9 (94.3%)	17,273.4 (96.2%)
less than 20	603.4 (3.6%)	71.4 (5.7%)	674.8 (3.8%)
Residence			

Urban	4,851.8 (29.0%)	250.3 (20.1%)	5,102.1 (28.4%)
Rural	11,852.2 (71.0%)	993.9 (79.9%)	12,846.1 (71.6%)
Mother education			
a education	13,472.5 (80.7%)	958 (77.0%)	14,430.5 (80.4%)
b no education	3,231.5 (19.3%)	286.2 (23.0%)	3,517.7 (19.6%)
Partner education			
a education	15,311.1 (91.7%)	1,106.1 (88.9%)	16,417.2 (91.5%)
b no education	1,392.9 (8.3%)	138.2 (11.1%)	1,531 (8.5%)
Mother occupation			
a other	3,999.5 (23.9%)	254.8 (20.5%)	4,254.4 (23.7%)
c Household, unskilled manual, not working	5,307.9 (31.8%)	350 (28.1%)	5,657.9 (31.5%)
d Agriculture	7,396.6 (44.3%)	639.4 (51.4%)	8,036 (44.8%)
Partner occupation			
a other	7,484.6 (44.8%)	455.9 (36.6%)	7,940.5 (44.2%)
c Household, unskilled manual, not working	2,704.3 (16.2%)	223 (17.9%)	2,927.3 (16.3%)
d Agriculture	6,515.1 (39.0%)	565.3 (45.4%)	7,080.4 (39.4%)

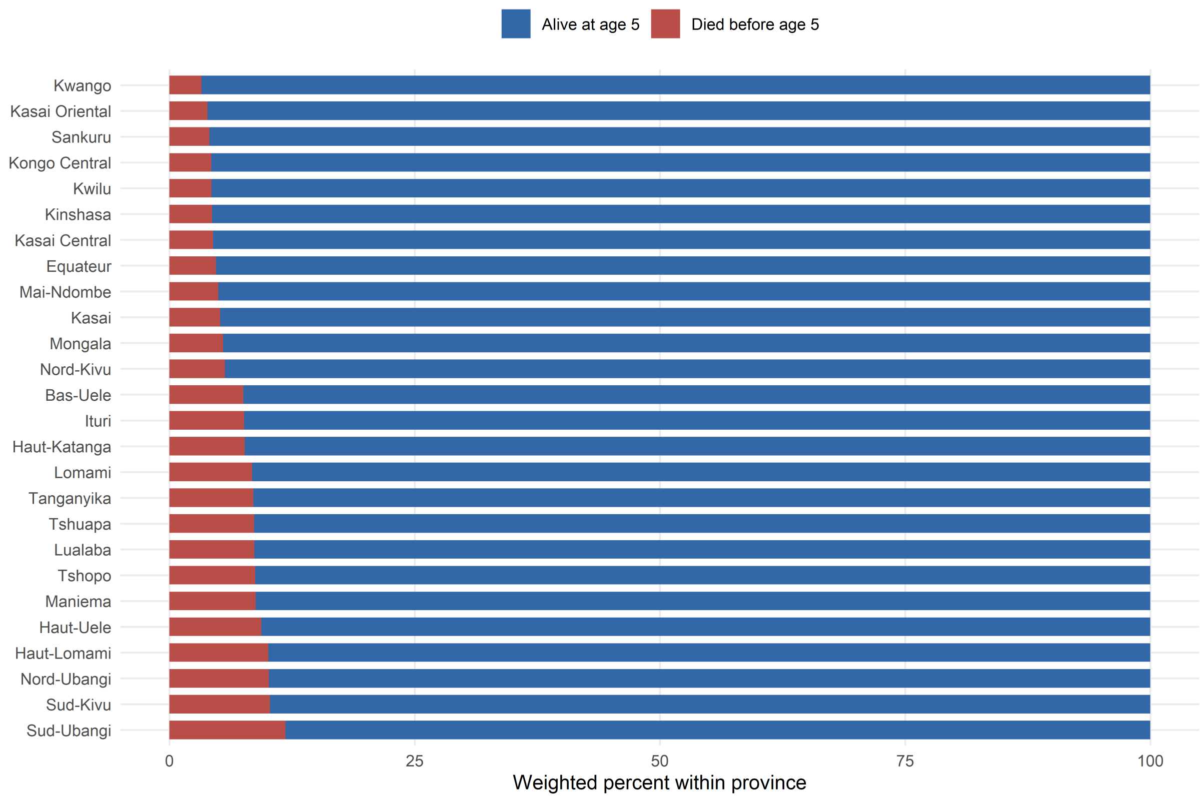

Figure 7: Weighted province distribution by under-five mortality status.

2.1.2 Appendix R2: Regression and Predictive-Forest Details

Table 5: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.983	0.268	-11.135	< 0.001	-3.509	-2.458

Low-skill HH	0.009	0.241	0.037	0.97074	-0.463	0.480
Child sex	0.306	0.085	3.591	< 0.001	0.139	0.472
Birth group: b 2-4 short	-0.003	0.144	-0.023	0.98128	-0.285	0.278
Birth group: c 2-4 long	-0.553	0.133	-4.153	< 0.001	-0.814	-0.292
Birth group: d 5+ short	0.434	0.146	2.980	0.00289	0.149	0.720
Birth group: e 5+ long	-0.372	0.136	-2.726	0.00642	-0.639	-0.104
Mother age	0.319	0.178	1.787	0.07393	-0.031	0.668
Residence	0.384	0.137	2.799	0.00514	0.115	0.654
Mother education	0.039	0.113	0.345	0.73043	-0.183	0.261
Partner education	0.199	0.151	1.319	0.18708	-0.097	0.496
Mother occupation: c Household, unskilled manual, not working	-0.087	0.223	-0.390	0.69656	-0.523	0.349
Mother occupation: d Agriculture	0.051	0.131	0.391	0.69544	-0.206	0.309
Partner occupation: c Household, unskilled manual, not working	0.151	0.177	0.854	0.39300	-0.196	0.498
Partner occupation: d Agriculture	0.103	0.105	0.980	0.32707	-0.103	0.309
Province: Kwango	-0.707	0.420	-1.684	0.09218	-1.530	0.116
Province: Kwilu	-0.540	0.317	-1.702	0.08886	-1.162	0.082
Province: Mai-Ndombe	-0.335	0.313	-1.070	0.28473	-0.949	0.279
Province: Kongo Central	-0.420	0.454	-0.926	0.35445	-1.310	0.469
Province: Equateur	-0.326	0.336	-0.972	0.33098	-0.984	0.331
Province: Mongala	-0.273	0.312	-0.877	0.38059	-0.884	0.338
Province: Nord-Ubangi	0.302	0.302	0.999	0.31797	-0.291	0.894
Province: Sud-Ubangi	0.489	0.336	1.457	0.14506	-0.169	1.147
Province: Tshuapa	0.176	0.321	0.549	0.58318	-0.453	0.806
Province: Kasai	-0.303	0.335	-0.906	0.36519	-0.960	0.353
Province: Kasai Central	-0.455	0.397	-1.146	0.25179	-1.232	0.323
Province: Kasai Oriental	-0.615	0.606	-1.016	0.30961	-1.803	0.572
Province: Lomami	0.133	0.321	0.414	0.67890	-0.496	0.761
Province: Sankuru	-0.646	0.335	-1.930	0.05366	-1.302	0.010
Province: Haut-Katanga	-0.002	0.315	-0.007	0.99429	-0.619	0.615
Province: Haut-Lomami	0.368	0.285	1.288	0.19765	-0.192	0.927
Province: Lualaba	0.377	0.297	1.267	0.20508	-0.206	0.959
Province: Tanganyika	0.154	0.292	0.528	0.59752	-0.419	0.727
Province: Maniema	0.172	0.334	0.516	0.60586	-0.482	0.826
Province: Nord-Kivu	-0.368	0.347	-1.058	0.29019	-1.049	0.314
Province: Bas-Uele	0.137	0.291	0.470	0.63804	-0.433	0.707
Province: Haut-Uele	0.197	0.323	0.611	0.54116	-0.436	0.831
Province: Ituri	0.232	0.311	0.746	0.45549	-0.378	0.842
Province: Tshopo	0.186	0.328	0.566	0.57124	-0.457	0.829
Province: Sud-Kivu	0.335	0.292	1.146	0.25199	-0.238	0.908

Table 6: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
mn_log_loss	1	150	0.2568	0.0038	5	1
mn_log_loss	1	80	0.2568	0.0038	5	2
mn_log_loss	1	10	0.2568	0.0038	5	3
mn_log_loss	15	150	0.2606	0.0044	5	4
mn_log_loss	29	150	0.2630	0.0045	5	5
mn_log_loss	43	150	0.2643	0.0046	5	6
mn_log_loss	15	80	0.2656	0.0047	5	7

mn_log_loss	29	80	0.2693	0.0051	5	8
mn_log_loss	43	80	0.2781	0.0086	5	9
mn_log_loss	15	10	0.2964	0.0058	5	10
mn_log_loss	29	10	0.3650	0.0074	5	11
mn_log_loss	43	10	0.4987	0.0208	5	12
roc_auc	1	80	0.6270	0.0105	5	1
roc_auc	1	10	0.6269	0.0103	5	2
roc_auc	1	150	0.6262	0.0110	5	3
roc_auc	15	150	0.6103	0.0068	5	4
roc_auc	29	150	0.6066	0.0077	5	5
roc_auc	43	150	0.6055	0.0081	5	6
roc_auc	15	80	0.6044	0.0074	5	7
roc_auc	29	80	0.6004	0.0086	5	8
roc_auc	43	80	0.5972	0.0093	5	9
roc_auc	15	10	0.5824	0.0068	5	10
roc_auc	29	10	0.5738	0.0072	5	11
roc_auc	43	10	0.5702	0.0076	5	12

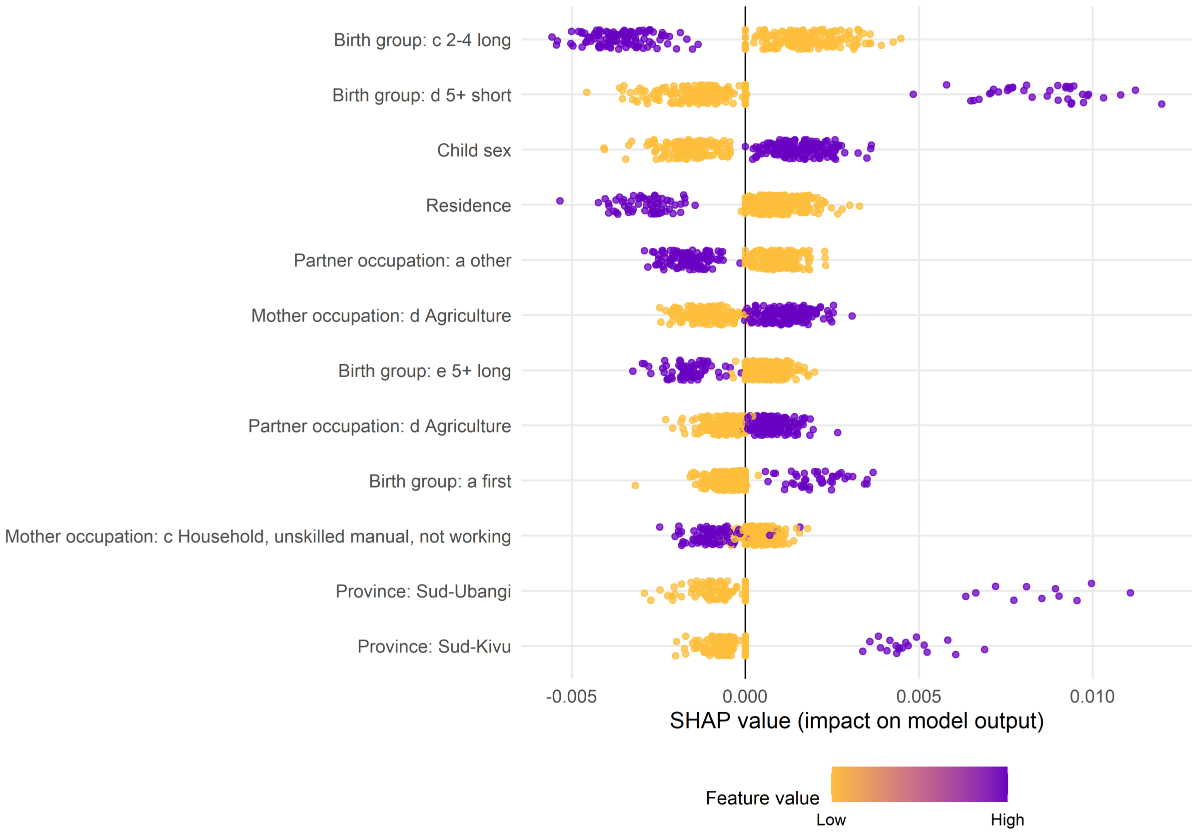

Figure 8: SHAP summary plot for the predictive ranger model.

2.1.3 Appendix R3: Concentration-Index Tree Details

Table 7: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain/root	Terminal nodes

CI	0.1112	0.0405	0.3646	NA	2
CIg	0.0077	0.0050	0.6549	NA	7
CIc	0.0308	0.0202	0.6549	NA	7
L	0.0818	0.0781	0.9542	NA	7

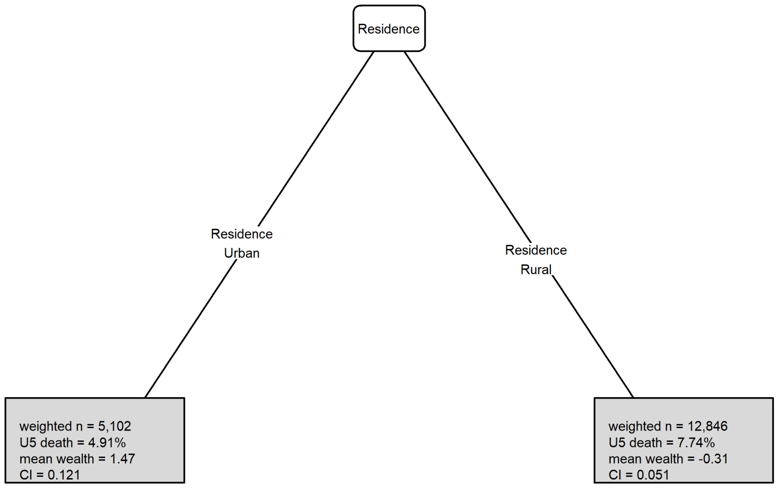

Figure 9: Concentration-index tree fitted with the CI criterion.

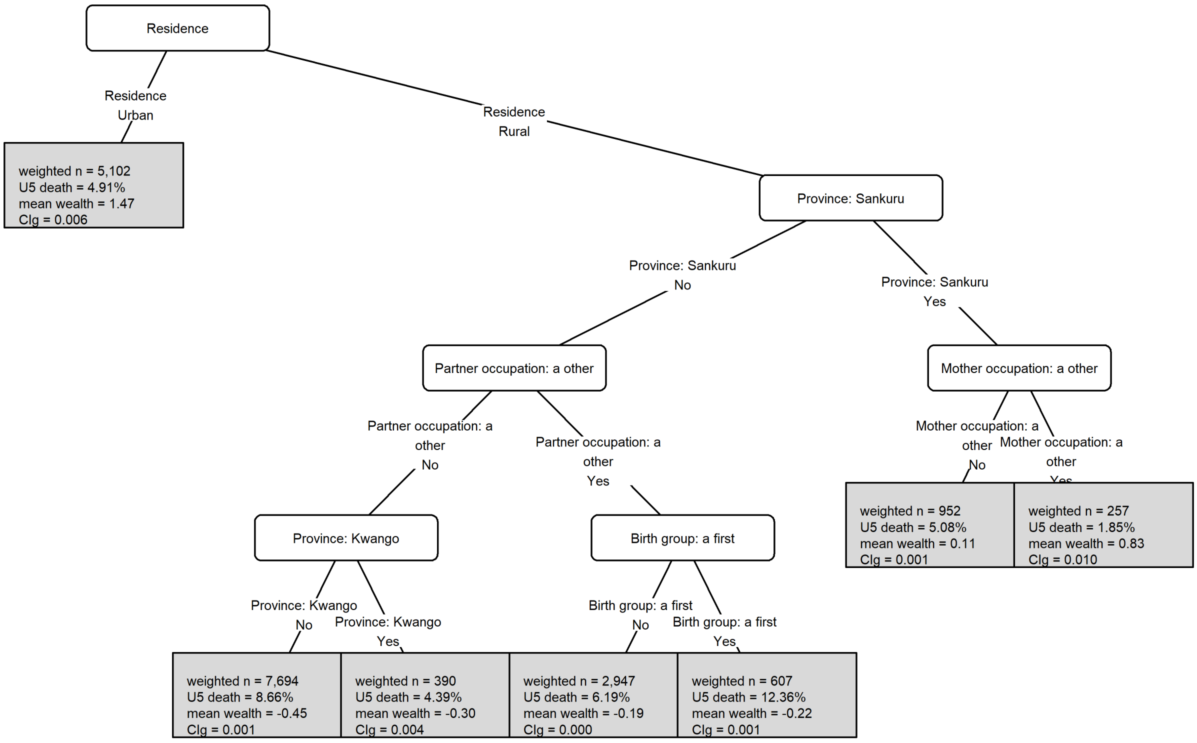

Figure 10: Concentration-index tree fitted with the CIg criterion.

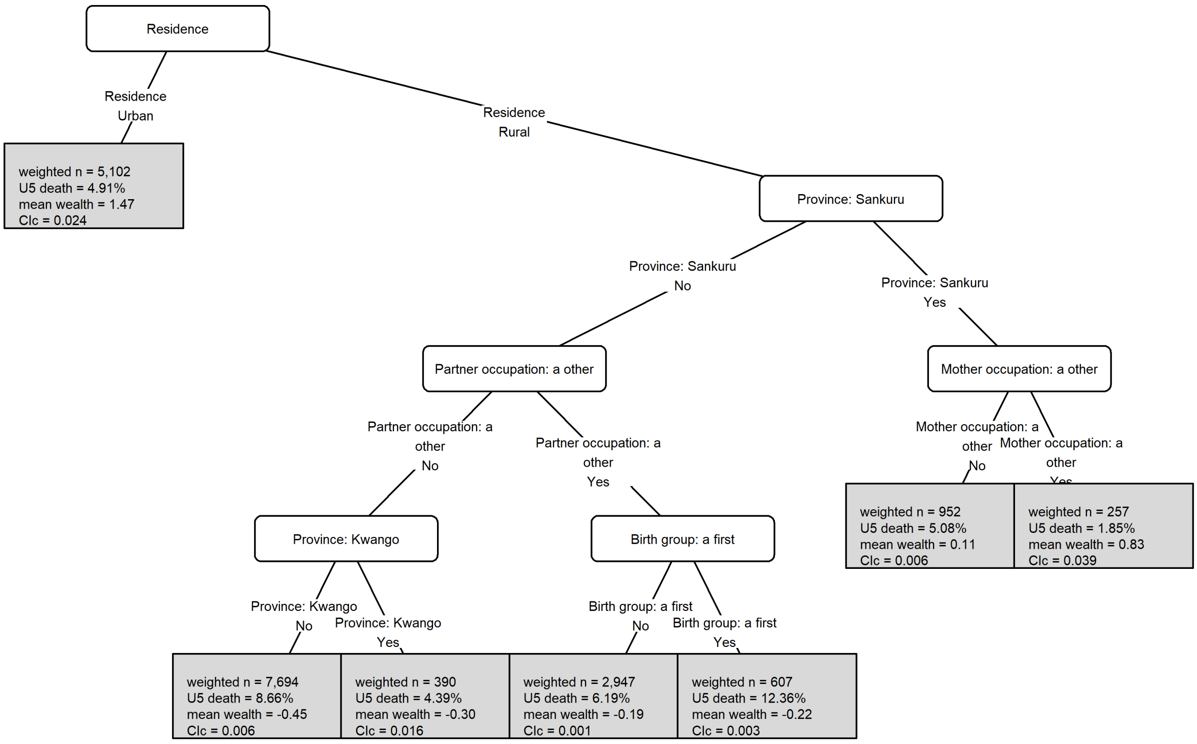

Figure 11: Concentration-index tree fitted with the CIc criterion.

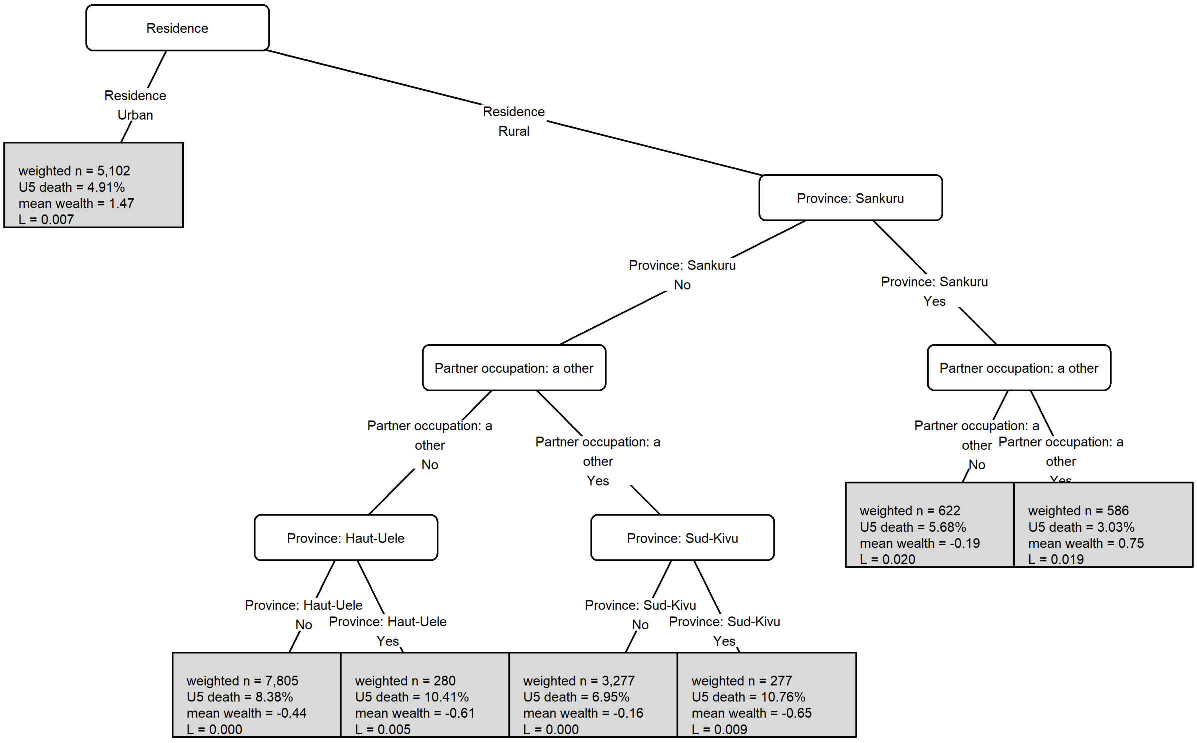

Figure 12: Concentration-index tree fitted with the L criterion.

2.1.4 Appendix R4: Concentration-Index Forest Details

Table 8: Selection metrics for fitted concentration-index forests

Criterion	Trees	mtry	Root objective	Train gain	Train gain/root	Validation gain/root	Mean termina
CI	100	6	0.1112	0.0092	0.0825	NA	
CIg	100	6	0.0077	0.0010	0.1302	NA	
CIc	100	6	0.0308	0.0035	0.1140	NA	
L	100	6	0.0818	0.0390	0.4760	NA	

Table 9: Forest variable importance by concentration-index criterion

criterion	rank	variable	importance
CI	1	Residence	0.0077
CI	2	Partner occupation: a other	0.0019
CI	3	Province: Kinshasa	0.0007
CI	4	Mother occupation: c Household, unskilled manual, not working	0.0004
CI	5	Mother occupation: a other	0.0002
CI	6	Mother occupation: d Agriculture	0.0001
CI	7	Province: Sankuru	0.0001
CI	8	Birth group: b 2-4 short	0.0000
CIg	1	Residence	0.0005
CIg	2	Mother occupation: d Agriculture	0.0002
CIg	3	Partner occupation: a other	0.0002

CIg	4	Province: Kinshasa	0.0001
CIg	5	Partner occupation: d Agriculture	0.0001
CIg	6	Province: Sankuru	0.0000
CIg	7	Mother occupation: a other	0.0000
CIg	8	Province: Tshopo	0.0000
CIg	9	Child sex	0.0000
CIg	10	Province: Kwango	0.0000
CIg	11	Birth group: d 5+ short	0.0000
CIc	1	Residence	0.0020
CIc	2	Partner occupation: a other	0.0007
CIc	3	Mother occupation: d Agriculture	0.0007
CIc	4	Province: Kinshasa	0.0003
CIc	5	Partner occupation: d Agriculture	0.0002
CIc	6	Province: Sankuru	0.0001
CIc	7	Province: Ituri	0.0000
CIc	8	Child sex	0.0000
L	1	Residence	0.0107
L	2	Mother occupation: d Agriculture	0.0088
L	3	Partner occupation: d Agriculture	0.0083
L	4	Mother education	0.0057
L	5	Partner occupation: a other	0.0049
L	6	Province: Sud-Kivu	0.0010
L	7	Partner education	0.0009
L	8	Province: Sankuru	0.0003
L	9	Low-skill HH	0.0001
L	10	Birth group: b 2-4 short	0.0000
L	11	Birth group: e 5+ long	0.0000

Table 10: Surrogate-tree variable importance for the selected concentration-index forest

rank	**variable**	**importance**
1	Residence	0.0176
2	Partner occupation: a other	0.0057
3	Mother occupation: d Agriculture	0.0008
4	Province: Sankuru	0.0005
5	Partner occupation: d Agriculture	0.0003
6	Mother education	0.0001
7	Partner occupation: c Household, unskilled manual, not working	0.0000

2.1.5 Appendix R5: Implementation Status

Implemented in the report: selected concentration-index tree and selected forest/surrogate-tree variable-importance plots in the results section; tree selection metrics, forest training-screen metrics, ranger cross-validation, SHAP summary plot, forest variable importance, and surrogate-tree variable importance in the appendix.

To implement for the final analysis run: rerun the generator with `RUN_CV=true` to populate cross-validated tree validation columns, and run forest tuning if cross-validated concentration-index forest validation metrics are needed rather than the current fixed-control training-screen metrics.